Response to Reviewer #2 (Round 2) — Manuscript CSEM-D-25-02344R1

We thank the reviewer for the careful re-reading of the revision and for identifying several points of internal consistency that needed reconciling. We address each in turn.

Major Points

1. On the revised Sharpe ratio (0.27 → 0.65)

We appreciate the reviewer pressing us for more specificity here, and we owe two clarifications.

What we can and cannot reconstruct. The correction from the original submission's SPY/UCB Sharpe ratio of 0.27 was made during an earlier revision pass, prior to the code refactor and audit trail we built for this round of review. The repository's preserved history begins after that fix was already applied, so we cannot produce a line-by-line diff against the pre-fix code the way we can for issues found during this round's audit (below). We can, however, describe the substance of the fix from the authors' direct recollection of the change, since one of us wrote both versions. The prediction-to-return conversion expresses each model's forecast as a return relative to the last *actual* observed price, $(\hat{y}_t - y_{t-1})/y_{t-1}$, exactly as implemented today (`compute_returns_from_preds` in our current codebase). In the pre-fix version, the reference point was instead the model's own *previous prediction* rather than the previous true price, i.e. effectively $(\hat{y}_t - \hat{y}_{t-1})/\hat{y}_{t-1}$. Because consecutive predictions from the same smoothly-trained model tend to sit close to one another, this framing suppressed the very signal the return was meant to capture, systematically understating the model pool's predictive edge relative to computing the return against the actual last price. Combined with (b) the introduction of the variance-aware UCB/Thompson Sampling policies described in our response to Comment 1, which materially changed the reported numbers because the original 0.27 figure was measured under a Softmax-only comparison, this return-computation fix accounts for the reported jump from 0.27 to the current values. We flag this as the authors' recollection rather than a recovered diff, since the pre-fix code itself no longer exists to verify against.

A separate, independently-found correction. During this revision we also identified and fixed an unrelated bug: our annualization factor was hardcoded at 252 trading days/year for every asset. For SPY this makes negligible difference (SPY trades ~251.5 bars/year in our sample, essentially 252 — Sharpe changes by well under 0.1%), but it materially inflates the Sharpe ratio for BTC-USD (~365 bars/year, no exchange holidays) and EURUSD=X (~260 bars/year). We disclose this separately because it is a distinct issue from the pct-change fix above, and because it is fully reproducible from our current codebase and test suite.

Resolving our own earlier inconsistency. We also note, on re-reading our first-round response, that we stated two different "resolved" Sharpe values (0.65 and 0.60) in the same section — an error on our part. For the record, Table 1 as it stands in this revision reports SPY/UCB Sharpe as **0.66** (heterogeneous pool) and **0.69** (homogeneous pool).

Disentangling the driver. The reviewer also asks us to separate the contribution of the calculation fix from the contribution of policy choice. We can do this partially using numbers already in Table 1: comparing Softmax (the policy used in the original submission) to UCB under the *current, corrected* pipeline isolates the policy-choice contribution alone. For SPY, homogeneous pool: Softmax Sharpe = 0.66, UCB Sharpe = 0.69 — a modest but real gap, smaller than the full 0.27→0.65 jump, which confirms the calculation fix (not recoverable in exact magnitude, as noted above) was the larger contributor. We now report a formal Diebold-Mariano test of whether this and other CMAB-vs-baseline gaps are statistically distinguishable from zero (see our response to your Comment/Additional Point 2 below and new Section 4.4).

Did the same error affect other original-submission statistics, including the L1/L2/BCE reward- function comparison? That comparison table no longer appears in the current manuscript at all — we address this directly under Additional Point 1 below, since it is the same underlying issue (a shift in the paper's central framing, not a residual of the pct-change bug).

2. On the consistency of the trading strategy (long-only vs. long–short)

The reviewer is correct, and we apologize for the confusion: our previous response's language ("zero-beta Long/Short framework," "exploratory short positions," "Long/Short CMAB") was simply wrong and does not describe our actual implementation. The strategy has always been, and remains, long/cash exactly as defined in Section 3.5 of the manuscript: $S_t \in \{0, 1\}$, with $S_t = 0$ liquidating fully into a zero-return cash position — never a short position. We have not changed the manuscript's implementation description, since it was already correct; we retract the erroneous framing from our prior response letter. This also closes out the third candidate driver the reviewer names at the end of this point: a short-selling/"position-timing" mechanism was never a real contributor to the Sharpe-ratio improvement, since the strategy never took short positions in the first place — it was an artifact of the prior letter's incorrect description, not an actual third driver alongside the calculation fix and policy choice discussed under Major Point 1.

3. On the exclusion of a risk metric from the context vector

The reviewer is correct on the theoretical point, and we have corrected the manuscript's wording accordingly: in a disjoint (per-arm) linear model such as ours, a feature shared across arms is *not* automatically non-discriminatory, since each arm learns its own weight vector and could in principle assign that shared input a different coefficient. Our original phrasing ("merely introduce a global, non-discriminatory bias term") overstated this. We have revised Section 3.4 to instead ground the design choice in the fact that such a feature reflects a property of the *asset*, not of the specific model producing the forecast, and therefore carries no genuinely arm-specific information – a narrower and more defensible claim than the one we originally made, and one that motivated us to test it empirically rather than rely on the argument alone.

We have run the empirical comparison the reviewer suggests: appending a shared, rolling realized-volatility feature (a 20-day trailing standard deviation of the target's own return series, identical across every arm at a given timestep) to the context vector for SPY, homogeneous pool, all three policies (5 seeds each, compared against a variance-matched 5-seed subsample of the real, already-published 25-seed baseline).

The result is more informative than a simple confirmation: it is **policy-dependent, not uniformly neutral**. For Softmax the feature is essentially inert (Sharpe 0.682 -> 0.679, within 5-seed noise). For Thompson Sampling it is mildly positive (0.666 -> 0.686). For LinUCB – the policy we identify as the strongest point-estimate performer in Table 1 – it is clearly detrimental (Sharpe 0.686 -> 0.642, a 6.4% relative decline, with annualized return falling correspondingly). We offer a plausible mechanism in the manuscript: LinUCB's exploration bonus is a function of each arm's estimated context covariance, and a fourth, non-discriminating dimension enlarges that estimate without contributing signal, inflating exploration noise – a class of harm confidence-bound methods are structurally more exposed to than Softmax (whose exploration scale depends on logit magnitude, not context covariance) or Thompson Sampling.

We therefore read this ablation as **strengthening** rather than merely confirming our theoretical argument: excluding the shared risk feature is not just well-motivated in principle but empirically consequential for our best-performing policy. We have added a one-sentence pointer to Section 3.4 of the manuscript, alongside the existing theoretical argument, with the full empirical comparison reported in new Appendix C. We are transparent about scope: this check uses 5 seeds on SPY only (not the full 25-seed, 5-asset design), given the wall-clock cost of the bandit stage at this scale on local hardware; extending it is left to the broader retraining pass discussed in our Conclusion.

Responses That Are Largely Satisfactory

Comment 1 (UCB "dominant"). We agree the margins are narrow for several assets and have softened this language (now "the point-estimate leader" rather than "definitively dominant"). We now tie the claim directly to the formal significance results in new Section 4.4: across all 30 configurations underlying Table 1 (5 tickers x 3 policies x 2 pool types), a paired Diebold-Mariano test finds **zero** configurations remain significant at alpha=0.05 after either Holm (family-wise) or Benjamini-Hochberg (false-discovery-rate) correction. Only one configuration has a raw, uncorrected p<0.05 (GC=F/Thompson/heterogeneous, p=0.018; Holm-corrected p=0.551). A separate, less conservative block-bootstrap Sharpe-difference confidence interval excludes zero for 2 of 30 configurations (BTC-USD/Thompson/homogeneous and GC=F/Thompson/heterogeneous), both favoring CMAB. 20 of the 30 point estimates favor CMAB overall (the 10 that do not are concentrated in TLT, the asset we already flag as an exception). We now state this plainly in the text: UCB's lead is a consistent directional pattern across asset classes, not a set of individually confirmed per-asset effects.

Comment 2, first part (SGA/response coefficient). We have added a sentence to Section 2.1.2 acknowledging that the Softmax/SGA formulation's exploration scale is governed by logit magnitude rather than an explicit uncertainty estimate, and linking this directly to our motivation for including UCB and Thompson Sampling.

Comment 4 (placebo test dispersion). We have added a sentence to Section 4.3 stating explicitly that the reported placebo-test result reflects a single synthetic realization (fixed generator seed). Characterizing seed-to-seed dispersion requires retraining the full predictor pool per realization, which we are deferring to the broader retraining pass discussed in our Conclusion's Future Work paragraph (see also our response to Reviewer #3, Comment 1, regarding the compute cost of retraining).

Additional Points

1. Treatment of the original central claim

We agree this shift in emphasis should be acknowledged explicitly rather than left implicit. Because the manuscript itself should read as a self-contained document rather than a diff against its own submission history, we have not phrased this shift as an explicit comparison to an earlier draft within the manuscript text. Instead, the Conclusion now states directly, on its own terms, that beyond the short-term autocorrelation structure of the reward signal that motivates our context construction (Section 3.4), a central empirical finding of this study is that architectural heterogeneity within the predictor pool systematically penalizes static aggregation while CMAB-based selection thrives on it (Section 3.3). For the reviewer's benefit here in this letter: the original submission's central claim emphasized the reward-signal autocorrelation finding in relative isolation, and this revision foregrounds the architectural-heterogeneity finding as the empirically more consequential result, without altering or superseding the original autocorrelation finding itself. The L1/L2/BCE reward-function comparison table from the original submission was not affected by any calculation error — it was removed because the paper's focus shifted, not superseded by a corrected version of itself.

2. Statistical significance

We have added new Section 4.4, reporting a paired Diebold-Mariano test and a block-bootstrap Sharpe-ratio-difference confidence interval for the CMAB-greedy vs. static-ensemble comparison, for every one of the 30 configurations in Table 1, computed from the same 25 independent seeds already used to produce the table. We apply both Holm (family-wise) and Benjamini-Hochberg (false discovery rate) corrections across the 30 tests.

The honest result: **0 of 30 configurations remain significant after either correction at alpha=0.05.** We report this plainly rather than searching for a more favorable test specification. 20 of the 30 point estimates favor CMAB (directionally consistent with Table 1, with the shortfall concentrated in TLT, our already-noted exception), and a less conservative block-bootstrap Sharpe-difference CI excludes zero for 2 of 30 configurations (both favoring CMAB) – but neither the raw DM p-values nor this bootstrap procedure survive family-wise or FDR correction across the full family of 30 tests. We believe this is the right, if humbling, way to answer the question: the paper's contribution is a consistent directional edge across diverse asset classes and architectures, not a set of individually statistically confirmed per-asset effects, and we have revised the relevant claims in Sections 4.1 and 5 (Conclusion) to reflect this precisely.

3. Internal consistency of stated values

Evaluation period. The current manuscript states a single, consistent evaluation period ("January 2000 to March 2026," Table 1 caption) that matches our current cached data exactly (verified: our SPY series' out-of-sample predictions run through 2026-03-16). The "January 2000 to January 2024" language the reviewer may be recalling comes from our first-round response letter, which was written against an earlier data pull; our corpus has since been refreshed and the manuscript's caption was updated to match, but we had not revised the earlier response letter's wording to match — we do so now. We have also clarified the caption itself: "January 2000" is the start of the raw data pull, while out-of-sample testing (and therefore every number in Table 1) only begins after the initial four-year walk-forward training window, i.e. from approximately 2004 onward — pre-empting a version of this exact ambiguity from recurring in a future round.

GARCH vs. EGARCH. The placebo test described in Section 4.3 of the manuscript is, and has consistently been, a **GARCH(1,1)** process — we have verified this directly against our synthetic-data generator, which implements the standard symmetric GARCH(1,1) variance recursion with no asymmetric/leverage term. The "EGARCH(1,1), Section 4.2" description that appeared in our response to Reviewer #3 in the previous round was an error in that letter — the underlying ticker label used internally for this dataset ("EGARCH") is a legacy naming artifact from early development and does not reflect the actual generative model. We regret the confusion and confirm GARCH(1,1)/Section 4.3 is the correct and only description.

We thank the reviewer again for the thorough second read and believe the manuscript is substantially strengthened by these clarifications and the new empirical analyses.